

RE3 - Stability of a differential amplifier with feedback: h-parameter solution

Data: $A_1 = 8000$, $R_{id} = 10 \text{ k}\Omega$, $R_o = 5 \text{ k}\Omega$, $R_1 = 40 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$. $R_L = 100 \text{ k}\Omega$ is **not** included in A.

a) Topology

Series-parallel (voltage-voltage) feedback. The output voltage is sampled and the input voltages are compared; the stabilized parameter is the voltage gain.

b) Matrices of A and β

$$H_A = [[R_{id}, 0], [-A_1 R_{id}/R_o, 1/R_o]] = [[10 \text{ k}\Omega, 0], [-16000, 2 \times 10^{-4} \text{ S}]]$$

$$H_\beta = [[R_1/R_2, R_2/(R_1+R_2)], [-R_2/(R_1+R_2), 1/(R_1+R_2)]] = [[8 \text{ k}\Omega, 0.2], [-0.2, 2 \times 10^{-5} \text{ S}]]$$

$$H = H_A + H_\beta = [[18000 \text{ }\Omega, 0.20], [-16000.2, 0.00022 \text{ S}]], \det H = 3204.00$$

c) Blocks A' and β'

$$A' = -h_{21}/(h_{11}h_{22}) \text{ (with H)} = -(-16000.2)/(18000 \times 0.00022) = \mathbf{4040.5}$$

$$\beta' = h_{12} = \mathbf{0.20}$$

$$A'\beta' = 808.1 \text{ (58.1 dB)}$$

d) Voltage gain and output impedance

$$A_f = V_o/V_i = -h_{21}/\det H = \mathbf{4.9938} \text{ (about } 1/\beta' = 5.00)$$

$$Z_i = \det H/h_{22} = \mathbf{14.56 \text{ M}\Omega}$$

$$Z_o = h_{11}/\det H = \mathbf{5.62 \text{ }\Omega} \text{ (PDF: } 5.6 \text{ }\Omega)$$

e) Dominant pole and phase margin

Pole from $f_{p2}/(A'\beta') = 10 \text{ MHz} / 808.1 = \mathbf{12.37 \text{ kHz}}$. With this pole the crossover is 7.86 MHz and PM = 51.9°.

Numerical solution for PM = 45°: $f_{p1}' = \mathbf{17.59 \text{ kHz}}$, crossover 10.04 MHz. The loop has two poles, so the phase reaches -180° only at infinity. The gain at 100 MHz gives GM = **37.0 dB**.

Notes

- Including R_L in A ($h_{22A} = 1/R_o + 1/R_L = 2.1 \times 10^{-4} \text{ S}$) gives $H_{22} = 2.3 \times 10^{-4} \text{ S}$ and **$A' = 3865$** , the value in RE3.pdf. The PDF's A' therefore includes R_L in A, which explains the difference from the 4040 found here with R_L excluded. R_L is not added to h_{22A} : the load is applied afterwards as a divider, $V_o/V_i \text{ (with } R_L) = A_f \cdot R_L/(R_L + Z_o) = 4.9938 \times 0.99994 = 4.9935$.

- The 134 dB gain quoted in e) is not consistent with $A_1 = 8000$ (78 dB). With the requested formula $f_{p2}/(A'\beta')$ and $A' = 4040$, the loop gain is 808 (58.1 dB).